

JEE MAIN REVISION MASTER: COMPLEX NUMBERS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 96 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Complex Numbers - Core Application	96 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Complex Numbers - Core Application (Very High Yield — 96 PYQs)

Core Principle: Key problem pattern in Complex Numbers.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "The number of values of $z \in \mathbb{C}$, satisfying the equations $|z-(4+8i)|=\sqrt{10}$ and $|z-(3+5i)|+|z-(5+11i)|=4\sqrt{5}$, is :..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Let $a, b \in \mathbb{R}$. Let α, β be the roots of the equation $x^2+ax+b=0$. If $\beta-\alpha=\sqrt{11}$ and $\beta^2-\alpha^2=3\sqrt{11}$..." [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "Let x and y be real numbers such that $50((2x/1+3i)-(y/1-2i))=31+17i, i=\sqrt{-1}$. Then the value of $10(x-3y)$ is :..." [Ans: D]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Morning Shift (Q#1)

Let $a, b \in \mathbb{R}$. Let α, β be the roots of the equation $x^2 + ax + b = 0$. If $\beta - \alpha = \sqrt{11}$ and $\beta^2 - \alpha^2 = 3i\sqrt{11}$, then $(\beta^3 - \alpha^3)^2$ is equal to:

- (A) 160
- (B) 176 [Correct]
- (C) 194
- (D) 187

Correct Answer: B

Solution Analysis: Given: α, β are roots of $x^2 + ax + b = 0$ $\beta - \alpha = \sqrt{11}$ $\beta^2 - \alpha^2 = 3i\sqrt{11}$ We have to find $(\beta^3 - \alpha^3)^2$. Now, $\beta^2 - \alpha^2 = (\beta - \alpha)(\beta + \alpha)$ Substituting the given values, $3i\sqrt{11} = \sqrt{11}(\beta + \alpha)$ Since $\sqrt{11} \neq 0$, we get $\beta + \alpha = 3i$ Now use the identity $\beta^3 - \alpha^3 = (\beta - \alpha)(\beta^2 + \alpha\beta + \alpha^2)$ So first let us find $\beta^2 + \alpha\beta + \alpha^2$. We know $(\beta + \alpha)^2 = \beta^2 + \alpha^2 + 2\alpha\beta$ Also,
$$\alpha\beta = \frac{(\alpha + \beta)^2 - (\alpha - \beta)^2}{4}$$